1. Clinical Question & Core Hypothesis

Clinical Question

Osteopontin (SPP1) is an independent prognostic signal but its association with favorable versus unfavorable outcomes appears context-dependent — what tumor microenvironmental or tumor-intrinsic factor(s) convert SPP1-associated biology from risk-driving to prognosis-favoring? 2) I built on the single provided mechanistic path linking an EPB41L2-associated risk milieu → elevated Osteopontin (SPP1) → prognosis modulation → FGFR3-associated favorable outcome; SPP1 and FGFR3 form the most direct nodes that could mechanistically flip SPP1 effects. 3) Supporting evidence (used as secondary rationale) is the upstream EPB41L2 node: cytoskeletal/membrane alterations linked to EPB41L2 plausibly shape tumor–stroma interactions and SPP1 production, providing an upstream modifier to test. 4) I selected overall survival (OS) because it is the most robust, least ambiguous clinical endpoint available in TCGA-BRCA for assessing net prognostic effect. 5) Deliberate exclusions: I excluded EPB41L2 from the core hypothesis to keep the mechanistic claim sharp (EPB41L2 is listed as a secondary modifier to be interrogated as an upstream regulator). I excluded imaging features tied to immune organization (TILs/TLS) because the path centers on matricellular/stromal signaling, making stromal-to-tumor ratio (TSR) the most biologically relevant imaging feature?

Working Hypothesis

High TSR augments stromal-derived SPP1 signaling that promotes ECM remodeling and invasion, but tumor-cell FGFR3 activation reprograms SPP1-driven programs toward differentiation and survival, reducing invasiveness. Clinically, SPP1-high tumors with high TSR have improved OS only when FGFR3 is also high; if FGFR3 is low, SPP1-high/TSR-high tumors exhibit poorer OS, with EPB41L2 acting as an upstream modifier that modulates the strength of these interactions.

Element	Value
Imaging feature	Stromal-to-tumor ratio (TSR)
Anchor gene(s)	SPP1, FGFR3
Primary endpoint	OS

2. Why This Gene — Not Others

Primary hypothesis gene(s)

SPP1: SPP1) is an independent prognostic signal but its association with favorable versus unfavorable outcomes appears context-dependent — what tumor microenvironmental or tumor-intrinsic factor(s) convert SPP1-associated biology from risk-driving to prognosis-favoring? 2) I built on the single provided mechanistic path linking an EPB41L2-associated risk milieu → elevated Osteopontin (SPP1) → prognosis modulation → FGFR3-associated favorable outcome; SPP1 and FGFR3 form the most direct nodes that could mechanistically flip SPP1 effects.

FGFR3: FGFR3-associated favorable outcome; SPP1 and FGFR3 form the most direct nodes that could mechanistically flip SPP1 effects.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core

Secondary	EPB41L2	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- EPB41L2: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.